NITTE EDUCATION TRUST | NITTE MEENAKSHI INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTION UNDER VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELGAUM) YELAHANKA, BANGALORE – 560 064

A Project Report on

“DESIGN AND ANALYSIS OF VARIOUS SWIRLER CONFIGURATION FOR NON -PREMIXED COMBUSTION”

Submitted in partial fulfilment for the course

Bachelor of Engineering

In

Department of Aeronautical Engineering

Academic Year: 2025-26

SUBJECT CODE: 22AEP77

Submitted by

VAISHNAVI KHUBA (1NT22AE107)

HARI HARAN V (1NT23AE404)

SUCHINTH K G (1NT23AE413)

Under the Guidance of

Dr. SRIKANTH H V

Professor & HOD

Dept. of Aeronautical Engineering

NMIT, Bangalore

KNOWLEDGE • CHARACTER • UNITY

NITTE EDUCATION TRUST | NITTE MEENAKSHI INSTITUTE OF TECHNOLOGY

Department of Aeronautical Engineering

Certificate

This is to certify that the Project Report entitled "**Design and Analysis of Various swirler configuration for a non-premixed combustion**" is a Bonafide work Carried out by **HARIHARAN V, (1NT23AE404), SUCHINTH KG (1NT23AE413) and VAISHNAVI KHUBA (1NT22AE107)** in partial fulfilment for the award of B.E Degree in **Aeronautical Engineering** of the **Visvesvaraya Technological University, Belgaum** during the academic year 2025-26. It is certified that all suggestions have been incorporated in the Report. The report has been approved as it satisfies the academic requirements in respect of Project work prescribed as per the Autonomous scheme of **Nitte Meenakshi Institute of Technology**, for the Degree of Aeronautical Engineering.

Dr. Srikanth H V
Professor& HOD
Dept. of Aeronautical Engineering
NMIT

Dr. Srikanth H V
Professor& HOD
Dept. of Aeronautical Engineering,
NMIT

Dr. H C Nagaraj
Principle, NMIT

Name of the Examiner   Signature with Date

1

2

ACKNOWLEDGEMENT

It is time to express our sincere gratitude to all who are responsible for the successful completion of our Project phase. Firstly, we would like to express our sincere regards and heart full thanks to our guide **Dr. Srikanth H V**, Professor & Head Department of Aeronautical Engineering, NMIT. For his constant encouragement, guidance and support and **Dr. Srikanth H V** Professor and Head of Aeronautical Engineering Department, NMIT.

We take this opportunity to express our sincere words of gratitude to our beloved Principal **Dr. HC Nagaraj** for all the support and encouragement during our undergraduate studies.

We express our deep and sincere gratitude to Professor **N. R Shetty**, Director, Visionary of NMIT, Bangalore, who is kind hearted and role model for all the youngsters with his achievements for providing us an opportunity to reach the goal in education.

Our sincere gratitude is due towards Management of our institution **Nitte Meenakshi Institute of Technology, Bangalore.** We are grateful to all the teaching and non-teaching staff of Department of Aerospace Engineering NMIT, Bangalore, for providing us support and required facilities.

Last but not the least, we would like to thank our Parents and Friends who have helped us directly and indirectly in all possible ways for our success.

DECLARATION

We hereby declare that the project entitled "**Design and Analysis of Various swirler configuration for a non-premixed combustion**" submitted by us to Nitte Meenakshi Institute of Technology, Bangalore in partial fulfilment of the requirement for the award of the degree of B.E in Aeronautical Engineering is a record of Bonafide project work carried out under the guidance from **Dr. Srikanth HV**, Professor & Head, Department of Aeronautical Engineering, Bengaluru. This work is not previously the basis for award of any degree nor has been submitted elsewhere for the award of any degree.

PLACE: BANGALORE

DATE: 17/11/2025

HARIHARAN V
1NT23AE404

SUCHINTH K G
1NT23AE413

VAISHNAVI KHUBA
1NT22AE107

ABSTRACT

Swirlers play a vital role in stabilizing flames, enhancing mixing, and improving overall combustion efficiency in non-premixed combustion systems. In such systems, fuel and air enter the combustor through separate passages, and effective mixing must occur within the combustion zone to achieve complete and stable combustion. This study focuses on the design and analysis of various swirler configurations to evaluate their influence on flow behaviour, mixing characteristics, and combustion performance under non-premixed conditions. Different swirler geometries—including axial vane swirlers, radial swirlers, and hybrid configurations—were modelled and examined to understand how vane angle, vane count, and hub-to-tip ratio affect swirl strength and recirculation zone formation.

Using computational fluid dynamics (CFD), each swirler design was analysed with respect to velocity distribution, turbulent mixing, temperature fields, and species concentration during combustion. Particular attention was given to the formation of the central recirculation zone (CRZ), which is essential for flame anchoring in non-premixed systems. The study also assessed fuel–air interaction quality, pressure drop across the swirler, and combustion stability. Results show that swirl intensity significantly impacts flame structure, CO_2 formation, fuel consumption rate, and temperature uniformity. Axial swirlers provided balanced mixing with moderate pressure loss, while radial swirlers generated stronger turbulence at the cost of higher pressure drop. Hybrid swirlers exhibited improved flame stability and enhanced local mixing due to combined flow redirection effects.

Overall, the investigation highlights the importance of swirler geometry in determining combustion behaviour in non-premixed systems and provides valuable insights for designing low-emission, high-efficiency combustors.

TABLE OF CONTENTS

Sl. No.		Page. No.
i.	Acknowledgement	**iii**
ii.	Declaration	**iv**
iii.	Abstract	**V**
CHAPTER 1: INTRODUCTION		**01-04**
1.1	Role Of Swirl in Non-Premixed Flames	**01**
1.2	Importance Of Swirler Configuration in Diffusion Combustion	**01-02**
1.3	Flame Stabilization Mechanisms	**02-03**
1.4	Influence On Combustion Efficiency and Emissions	**03**
1.5	Fundamentals Of Swirl and Combustion	**05-09**
CHAPTER 2: LITERATURE SURVEY		**06-09**
CHAPTER 3: METHODOLOGY		**10-11**
3.1	Design Of Swirler Configurations	**10**
3.2	Cfd Simulation and Flow Analysis	**11**
CHAPTER 4: FORMULA AND CALCULATIONS		**12-15**
4.1	Procedure Of Calculating Axial Swirler Mass Flow	**12-13**
4.2	Calculation For Swirler Area	**13-14**
4.3	Swirler Sizing	**14-15**
CHAPTER 5: GEOMETRY DESIGN WITH COMBUSTION CHAMBER		**16-21**
5.1	Axial Curved Type Swirler	**16**
5.2	Axial Flat Type Swirler	**16**
5.3	Radial Type Swirler	**17**

5.4	Radial Type Swirler with Combustion Chamber	**17-18**
5.5	Axial Curved Type with Combustion Chamber	**18-19**
5.6	Axial Flat Type Swirler with Combustion Chamber	**20-21**
CHAPTER 6: RESULTS AND OBSERVATION		**22-30**
6.1	Co_2 Mass Fraction Contour	**22**
6.2	O_2 Mass Fraction Contour	**23**
6.3	Pressure Contour	**24**
6.4	N_2 Mass Fraction Contour	**25**
6.5	$C_{12}h_{23}$ Mass Fraction Contour	**26**
6.6	Temperature Contour	**27**
6.7	Velocity Contour	**28**
6.8	Static Temperature Graph	**29**
6.9	Static Pressure Graph	**30**
CHAPTER 7: CONCLUSION		**31**
CHAPTER 8: REFERENCES		**32-34**

LIST OF FIGURES

Fig. No.		Page. No.
4.1	Swirler Geometry Design.	**12**
5.1	Axial Curved Type Swirler.	**16**
5.2	Axial Flat Type Swirler.	**16**
5.3	Radial Type Swirler.	**17**
5.4(A)	Front And Side Radial Type Swirler with Combustion Chamber.	**17**
5.4(B)	Isometric View Radial Type Swirler with Combustion Chamber.	**18**
5.5(A)	Front And Side Axial Curved Type with Combustion Chamber.	**18**
5.5(B)	Isometric View Axial Curved Type with Combustion Chamber.	**19**
5.6(A)	Isometric View Axial Flat Type with Combustion Chamber.	**20**
5.6(B)	Front And Side Axial Flat Type with Combustion Chamber.	**21**
6.1	Co_2 Mass Fraction Contour	**22**
6.2	O_2 Mass Fraction Contour	**23**
6.3	Pressure Contour	**24**
6.4	N_2 Mass Fraction Contour	**25**
6.5	$C_{12}h_{23}$ Mass Fraction Contour	**26**
6.6	Temperature Contour	**27**
6.7	Velocity Contour	**28**
6.8	Static Temperature Graph	**29**
6.9	Static Pressure Graph	**30**

CHAPTER 1

INTRODUCTION

In non-premixed (diffusion) combustion systems, fuel and oxidizer are supplied separately and mixing occurs primarily due to turbulent diffusion within the combustion chamber. Unlike premixed combustion, where fuel and air are uniformly mixed prior to ignition, non-premixed combustion relies on the mixing process itself to control the rate of heat release and flame structure. This process is inherently controlled by fluid dynamic parameters such as turbulence intensity, recirculation, and flow stability—all of which can be profoundly influenced by swirling motion introduced through swirler devices.

1.1 Role of Swirl in Non-Premixed Flames

In diffusion flames, the stabilization of the flame front is a key challenge due to the limited mixing of reactants and the risk of blow-off at high velocities. Swirlers impart a strong tangential velocity component to the incoming air stream, generating a central recirculation zone (CRZ) downstream of the swirler. This CRZ serves as a continuous ignition source by entraining hot combustion products back into the incoming fuel-air region, ensuring sustained combustion even under lean conditions.

In non-premixed combustion, this swirl-induced recirculation not only enhances flame stability but also increases the residence time of reactants, improving combustion completeness and reducing unburned hydrocarbons (UHC). However, the extent of swirl must be optimized carefully—excessive swirl can lead to flame lift-off or excessive pressure losses, while insufficient swirl can cause poor mixing and unstable combustion.

1.2 Importance of Swirler Configuration in Diffusion Combustion

The performance of a non-premixed combustor is highly dependent on the geometry and configuration of the swirler. Swirler design determines the swirl strength, flow distribution, and the size of the recirculation zones. Several configurations are commonly employed:

- **Axial Swirlers:**

  Axial swirlers are composed of vanes mounted in the flow path at an angle to the axial direction. By varying vane angles (typically between 30° and 60°) and the number of vanes, designers can precisely control the swirl intensity and flow distribution. In non-premixed combustion, axial swirlers ensure good fuel-air interaction and help in anchoring diffusion flames close to the injector exit.

- **Radial Swirlers:**

  In radial swirlers, air is directed radially inward (or outward) before it exits axially into the combustion chamber. This creates a strong vortex and an intense mixing region ideal for heavy fuels or systems requiring enhanced atomization. These are often used in burners where the air must envelop the fuel jet for better diffusion-controlled combustion.

- **Tangential Swirlers:**

  Tangential swirlers introduce air through tangential slots or holes located on the burner periphery. The resulting swirling motion provides a high swirl number and a strong central vortex. Such swirlers are effective for large-scale diffusion flames, such as those in industrial furnaces and boiler burners.

1.3 Flame Stabilization Mechanisms

Swirlers aid flame stabilization in non-premixed systems through three primary mechanisms:

1. Recirculation of Hot Products: The central recirculation zone transports high-temperature gases upstream, promoting continuous ignition of incoming fuel-air mixtures.

2. Shear Layer Formation: Strong velocity gradients between the recirculating core and surrounding flow generate turbulence, improving mixing between fuel and air.

3. Pressure Gradient Effects: The adverse pressure gradient generated by swirl causes flow reversal near the centre line, forming a low-pressure core that anchors the flame.

The effectiveness of these mechanisms depends on the swirl number (S), a non-dimensional parameter defined as:

$$S = \frac{G_\theta}{RG_x}$$

where $G_\theta$is the axial flux of angular momentum, $G_x$is the axial flux of linear momentum, and Ris the characteristic radius. In non-premixed combustion, moderate swirl numbers (typically between 0.4 and 1.0) are found to provide the best balance between flame stability and pressure loss.

1.4 Influence on Combustion Efficiency and Emissions

The swirler configuration directly affects not only flame shape and stability but also combustion efficiency and pollutant formation:

- NOx Formation: High temperatures in the recirculation zone can lead to thermal NOx formation; however, improved mixing can reduce peak temperatures, thus lowering overall NOx emissions.
- CO and UHC Emissions: Properly designed swirlers enhance mixing and combustion completeness, minimizing CO and unburned hydrocarbons.
- Soot Formation: In non-premixed flames, swirl promotes rapid oxidation of soot precursors by improving oxygen availability near the flame zone.

1.5 Fundamentals of Swirl and Combustion

The concept of swirl in combustion systems dates back several decades. Gupta et al. (1984) defined swirl as the angular motion imparted to a flow, characterized by a swirl number (S) — the ratio of axial flux of angular momentum to the product of axial flux of axial momentum and characteristic radius. When the swirl number exceeds approximately 0.6, a strong internal recirculation zone (IRZ) is established, which stabilizes the flame and enhances mixing.

Swirler Types and Design Approaches

- **Axial Swirlers:**

  These consist of a set of vanes arranged at a certain angle to the axial direction, causing the air to rotate as it flows downstream. Axial swirlers are widely used in gas turbine combustors due to their compact design and controllable swirl intensity. Studies (e.g., Lefebvre & Ballal, 2010) show that vane angle, number of vanes, and hub-to-tip ratio significantly influence the swirl strength and pressure drop.

- **Radial and Tangential Swirlers:**

  Radial swirlers introduce air radially inward, whereas tangential swirlers impart rotation by directing air tangentially into the combustion chamber. Tangential swirlers generally provide a higher swirl number but can cause higher pressure losses.

- **Hybrid or Counter-Swirl Designs:**

  Some researchers have explored counter-rotating and multi-stage swirlers to improve mixing uniformity and control recirculation zones (Reddy et al., 2017). Counter-swirl designs can enhance flame stability and reduce emissions by improving fuel-air mixing.

CHAPTER 2

LITERATURE SURVEY

i. Design And Study on Performance of Axial Swirler for Annular Combustor by Changing Different Design Parameters.

AUTHORS: Bhupendra Khandelwal, Dong Lili, Vishal Sethi.

OBJECTIVE: To develop a systematic design methodology for axial swirlers by varying vane angle, vane number, and mass-flow conditions, and to analyse how these geometric parameters affect swirl strength, flow structure, and pressure drop inside an annular gas-turbine combustor.

METHODOLOGY: Designed several axial swirler configurations using pressure-loss and drag-coefficient methods. Used CFD simulations to study flow fields, pressure drop, turbulence patterns, and recirculation behavior. Compared different turbulence models to determine the most accurate one for swirling flows.

RESULTS: Swirler performance changed significantly with vane angle, vane count, and mass flow. Four distinct axial and radial velocity profiles were observed. Turbulence distribution consistently showed double peaks that weakened downstream. The study established a complete theoretical + empirical procedure to design practical axial swirlers.

ii. CFD Simulation Analysis of Non-Premixed Combustion Using a Novel Axial-Radial Combined Swirler for Emission Reduction Enhancement

AUTHORS: Nor Afzanizam Samiran, Khalil Firdaus Abu Mansor, Razlin Abd Rashid, Muhammad Suhail Sahul Hamid

OBJECTIVE: To evaluate a newly designed axial-radial combined swirler and determine whether combining swirl types can improve fuel–air mixing and reduce combustion emissions in non-premixed combustion.

METHODOLOGY: Designed axial and radial swirlers (each with 8 vanes) and a combined configuration in SolidWorks. Performed CFD simulations in ANSYS Fluent using standard k–ε turbulence model. Used LPG (30% propane, 70% butane) as fuel. Compared flow fields, temperatures, and emission outputs (CO, CO_2).

RESULTS: The combined swirler produced broader high-temperature regions and higher axial velocity.CO was reduced significantly due to more complete conversion to CO. The axial-only swirler produced slightly higher peak temperatures. Authors recommend increasing axial-vane inclination in the combined design to raise flame temperature if needed.

iii. Optimized Design of a Swirler for A Combustion Chamber of Non-Premixed Flame Using Genetic Algorithms

AUTHORS: Daniel Alejandro Zavaleta-Luna, Marco Osvaldo Vigueras-Zúñiga, Agustín L. Herrera-May, Sergio Aurelio Zamora-Castro, María Elena Tejeda-del-Cueto

OBJECTIVE: To optimize the design of an axial swirler using genetic algorithms with the goal of maximizing swirl number, producing strong recirculation, enhancing combustion efficiency, and reducing emissions (CO, NOx).

METHODOLOGY: Applied genetic algorithms to optimize vane angle, vane number, swirler area, and mass-flow-related parameters. Used CFD simulations employing the RNG k–ε model and mixture-fraction formulation for non-premixed combustion. Compared flow, temperature fields, and recirculation for baseline vs. optimized design.

RESULTS: The optimized swirler created a longer and more intense recirculation zone. Temperature distribution matched experimental measurements well, confirming model accuracy. Enhanced turbulence and mixing significantly reduced predicted CO and NOx formation. Shows GA-based optimization is effective for practical combustor design.

iv. Numerical And Experimental Investigation of a Non-Premixed Double Swirl Combustor

AUTHORS: Jiming Lin, Ming Bao, Feng Zhang, Yong Zhang, Jianhong Yang

OBJECTIVE: To study how outer-swirler vane angle and equivalence ratio influence flow behavior, combustor temperature, and NO emissions in a double-swirl non-premixed burner.

METHODOLOGY: Conducted experiments using a full-scale combustor with thermocouples and flue-gas analysis. Performed RANS CFD using realizable k–ε turbulence model and methane–air two-step chemistry. Tested multiple outer-swirler vane angles and varying tertiary-air equivalence ratios.

RESULTS: System showed no central recirculation zone due to draft-fan-driven flow distribution.NO emissions consistently decreased when outer-swirler blade angle increased. Increasing tertiary-air equivalence ratio boosted average furnace temperature by ~100 K and cut NO emissions by ~25%.Numerical results matched experimental measurements well.

v. Numerical Analysis of Flow Characteristics of a Swirl-Stabilized Premixed Burner with Different Swirl Vane Configurations

AUTHORS: Hae-ji Ju, Ju Hyeong Cho, Jeong Jae Hwang

OBJECTIVE: To compare flat-vane and curved-vane axial swirlers in a premixed burner and analyse their effects on pressure loss, mixing uniformity, NOx emissions, and jet penetration characteristics.

METHODOLOGY: Modelled both flat-vane and curved-vane axial swirlers in a premixed burner setup. Performed CFD simulations to study mixing quality, pressure drop, flame structure, and NOx formation. Analysed jet trajectory behavior under swirling crossflow conditions.

RESULTS: Curved vane reduced pressure loss by 35% by suppressing flow separation. NOx decreased by 69% due to more uniform mixing and lower local hotspots. Flat vanes maintained more uniform outlet velocity but caused higher NOx. Jet penetration prediction in swirling flow requires new models—conventional crossflow formulas do not apply.

vi. Experimental Analysis of Flow Through Rotating Combustion Swirler with Zero Degree Inlet and Outlet Angle of Guide Vane

AUTHOR: Mansha Kumari

OBJECTIVE: To experimentally study flow behavior, recirculation zone formation, and velocity/pressure distributions in a rotating 8-blade swirler with zero-degree guide-vane angles, using a five-hole probe.

METHODOLOGY: Designed a rotating 8-blade swirler (45° vane angle) with a guide vane stator at 0°/0°.Conducted experiments using a five-hole probe at several stations downstream of the diffuser. Measured recirculation size, swirl strength, and flow structure.

RESULTS: Flow exhibited strong swirl with clear central toroidal recirculation. Guide vanes at 0° reduced deformation and ensured stable flow entry. Data showed how swirl number directly influenced recirculation size and flow reversal. Provided complete characterization of recirculation and turbulence behavior for this swirler configuration.

vii. Research Gap

While many studies focus on single swirler configurations, limited research compares multiple swirler types under similar operating conditions. Moreover, the correlation between geometric parameters (vane angle, blade number, and hub ratio) and performance indicators (swirl number, pressure drop, and recirculation length) needs further exploration. This study addresses this gap by systematically designing and analysing various swirler configurations using both analytical and numerical methods.

viii. Objectives

The primary objective of this study is to design and evaluate multiple swirler configurations—axial, radial, and hybrid—specifically suited for non-premixed combustion systems. The work focuses on understanding how key geometric parameters such as vane angle, vane count, and overall vane geometry influence swirl intensity and the formation of the recirculation zone, which plays a crucial role in flame stabilization. Each swirler design will be examined for its ability to enhance fuel–air mixing efficiency, ensuring improved combustion quality. A comparison of pressure drop characteristics across the different swirlers will be performed to determine the most aerodynamically efficient configuration. Additionally, the study investigates combustor temperature distributions to identify potential hotspots and thermal gradients. Species concentration fields, including O_2, CO_2, N_2, and fuel mass fraction, will be measured to assess combustion completeness. The analysis also includes detailed examination of velocity fields and turbulent flow structures generated by each swirler design. Furthermore, the influence of swirler geometry on pollutant formation—including CO, NOx, and unburned fuel—will be evaluated. Ultimately, the study aims to identify the optimal swirler configuration that provides stable, highly efficient, and low-emission combustion within a non-premixed environment.

CHAPTER 3

METHODOLOGY

The research methodology involves two main stages:

- Design, Simulation (Analysis)
- Performance Evaluation.

3.1 Design of Swirler Configurations

- Software: SolidWorks / CATIA (for CAD modelling)
- Configurations Considered:
    1. Axial vane swirler (vane angles: 30°, 45°, 60°)
    2. Tangential inlet swirler
    3. Radial vane swirler
- Design Parameters:
    - Inlet diameter: 40 mm
    - Outlet diameter: 60 mm
    - Vane thickness: 2 mm
    - Number of vanes: 6, 8, 12
    - Hub-to-tip ratio: 0.5
    - Swirl number calculated using:

$$S = \frac{G_\theta}{R \times G_x}$$

where,

G_θ and G_x -axial fluxes of angular and axial momentum respectively.

3.2 CFD Simulation and Flow Analysis

- Software: ANSYS Fluent
- Solver Type: Steady-state, pressure-based, turbulent flow
- Turbulence Model: Realizable k–ε or k–ω SST
- Boundary Conditions:
  - Inlet: Uniform velocity (e.g., 10–30 m/s)
  - Outlet: Pressure outlet (0 Pa gauge)
  - Wall: No-slip condition
- Mesh: Unstructured tetrahedral mesh with refinement near walls and vane surfaces
- Grid Independence Test: Ensures accuracy of results

CHAPTER 4

FORMULA AND CALCULATION

4.1 PROCEDURE FOR CALCULATING AXIAL SWIRLER MASS FLOW

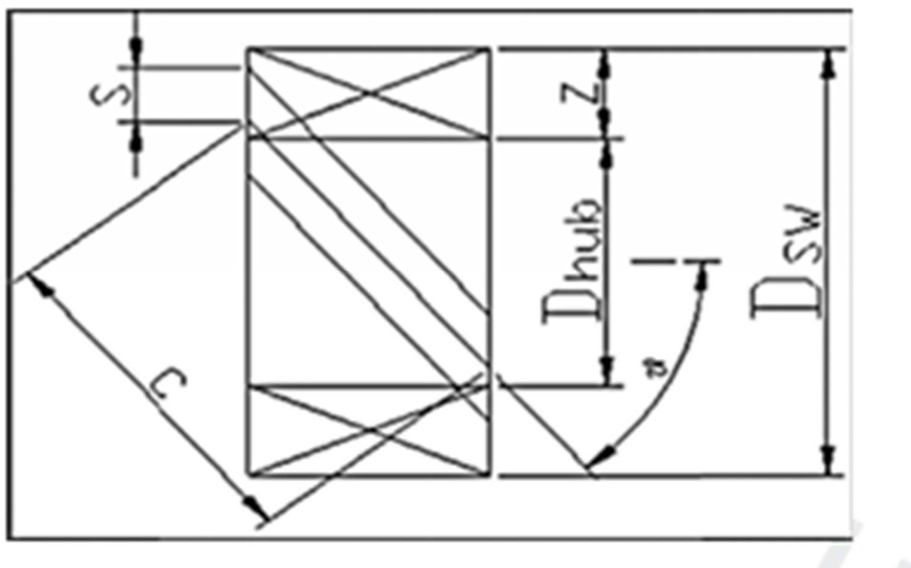

Fig 4.1 *Swirler Geometry Design.*

$$M_c = M_T * (1 - C_{cooling}) \quad \text{.....(1)}$$

Secondly the air to fuel ratio (AF) at the design point, cruise condition of turboprop engine is estimate during Eq. (2), which reflects the energy balance relationship of a typical gas turbine engine. In this equation, the specific heats Cpg and Cpa are found in thermodynamic properties table of air and have the value of 1210J/kg/K and 1093J/kg/K respectively. Fuel Calorific Value (FCV) is 43 MJ here and hence the air fuel ratio (AF) is 45.5.

$$AF = (FCV - Cpg\ TET)\ (Cpg\ TET - Cpa\ T3) \quad \text{.....(2)}$$

Thirdly the fuel mass flow (Mf) needs to be again calculated by using Eq. (3), although the fuel mass flow has been given in overall performance calculation, because here it is calculated from the prospective of combustion. The core mass flow (Mc) and air fuel ratio (AF) have been obtained by above two steps and so the fuel mass flow is 0.21kg/s.

$$Mf = Mc / AF \qquad \text{.....(3)}$$

Fourthly, some intermediate variables during calculation need be assumed because designing combustors cannot completely be theoretical; it is still said to be black art by many researchers . These assumed variables include ratio of dome cooling mass flow (Mcd) to core mass flow, Mcd/Mc, and ratio of air mass flow for atomizing fuel to fuel mass flow Mca/Mf if for air blast fuel jet. In order to ensure sufficient and effective atomization, the ratio of air mass flow for atomizing fuel to the fuel mass flow Mca/Mf has the value of 2–3, in this study value of 3 has been used. For annular combustors, the air for cooling dome is around 8% for older engines and 10–15% for modern engines, here the value of 15% is chosen .

Fifthly, the equivalence ratio in primary zone (ϕp) should be assumed in order to calculate swirling air mass flow (Mcs). For a conventional combustor, the equivalence ratio at primary zone ranges from 0.9 to 1.1. In this range there are relatively maximum flame temperatures and consequently the emission pollutants such as NOx may be observed. If using rich burn quick quench technology, the equivalence ratio is between 1.4 and 1.5. At this range the flame temperatures are lower than that of the conventional combustors and the level of emission are lower. Therefore, here the rich equivalence ratio is selected and has the value of 1.5.

Finally, once the equivalence ratio is selected in primary zone, the swirling air mass flow can be calculated by using Eq. (4). AF_{th} is the stoichiometric air–fuel ratio and the value is 14.7

$$Mcs = (AF_{th} / \phi p) \times Mf - Mca - Mcd \qquad \text{.....(4)}$$

4.2 CALCULATION OF SWIRLER AREA

The pressure loss coefficient method is used in this study to calculate the area of axial swirler. In general, pressure loss of swirler should be 60–70% of total pressure loss through combustor. Therefore, the pressure loss of swirler is here assumed as 65% of the total pressure loss through the combustor. Pressure drop coefficient of swirler $\Delta Psw / qref$

can be obtained by pressure loss of swirler over reference dynamic head. Pressure loss relationship of swirler is given in Eq. (5).

$$(\Delta Ps / qref) = k\,(Aref / As)^2 \sec^2\theta - (Aref / At)^2\,(Msw / M\,T) \qquad \text{.....(5)}$$

Vane shape factor k is 1.3 for flat vane and 1.15 for curved vane, which indicates that pressure loss of curved vane is less than that of flat vane swirler. In Eq. (5), all items except Aref/As are known, so Aref/As can be calculated and the area of swirler (As) can be obtained.

In typical combustors, swirl induced by recirculation is augmented by the airflow through the primary hole; a certain amount of air should be involved in mass flow of swirler for calculating area of swirler. Out of the quantity of airflow that enters in the primary zone of combustor through primary holes, 30–70% takes part in the recirculation . Consequently, the area of swirler that can be calculated from Eq. (5) need be corrected with augmented mass flow. Here allowance of 50% more airflow has been used.

4.3 SWIRLER SIZING

The flow area of swirler (Asw) is calculated by both pressure loss method and geometric structure. The area of swirler can be calculated by Eq. (6) if using the geometric dimensions.

$$Asw = (\pi / 4)\,(DT^2 - Dh^2) - 0.5n\delta\,(Dt - Dh) \qquad \text{.....(6)}$$

The hub diameter (Dh) is calculated by the outer diameter of fuel injector that has been roughly estimated to a value of 14 mm. The tip diameter of swirler (Dt) should be calculated from Eq. (6) and have the value of 28 mm. The hub tip ratio is the ratio of the hub diameter to the tip diameter and here is 0.5, which have better fall within the range of 1/3–2/3. The number of swirler vanes n commonly ranges from 8 to 16. The thickness of swirler vane is from 0.7 to 1.5 mm for the consideration of manufacturing. During designing, the thickness of vane for all swirlers is considered as 1 mm and the number of swirler vanes is different from each swirler. After calculating these dimensions, the swirl number SN need be checked by Eq. (7). If the swirl number SN is smaller than 1.0 then the number and the thickness need be changed repeatedly until the swirl number of around 1.0 is achieved. This is due to the reason that the swirl number for gas turbine combustors generally ranges from 1.0 to 1.5. For the annular swirler with constant vane angle θ, the expression is given by Eq. (7).

$$SN = (2 / 3) * [(1 - (Dh / Dt)^3) / (1 - (Dh / Dt)^2)] * \tan(\theta) \quad .....(7)$$

Dhub / Dsw is defined as hub tip ratio of swirler and the second item in above equation generally have the value between 1 and 1.5.

For the curved vane the expression is different from the constant vane swirler due to the different angles at trailing and leading edge of vane. Assuming the axial velocity component is uniform over cross section after swirler vane exit, the swirl number of curve vane swirler with vane exit angle θm can be represented as Eq. (8).

$$SN = (DT^2 + Dh^2) / (2\, DT^2) * \tan(\theta\, m) \quad .....(8)$$

The vane space, s, is calculated by Eq. (9):

$$s = \pi\, (Dt + Dh) / (2n) \quad .....(9)$$

The vane chord (c) can be obtained using Eq. (10) if assuming solid vane is (σ v)

$$c = \sigma\, v\, s \quad .....(10)$$

CHAPTER 5

DESIGN GEOMETRY

5.1 AXIAL CURVED TYPE SWIRLER

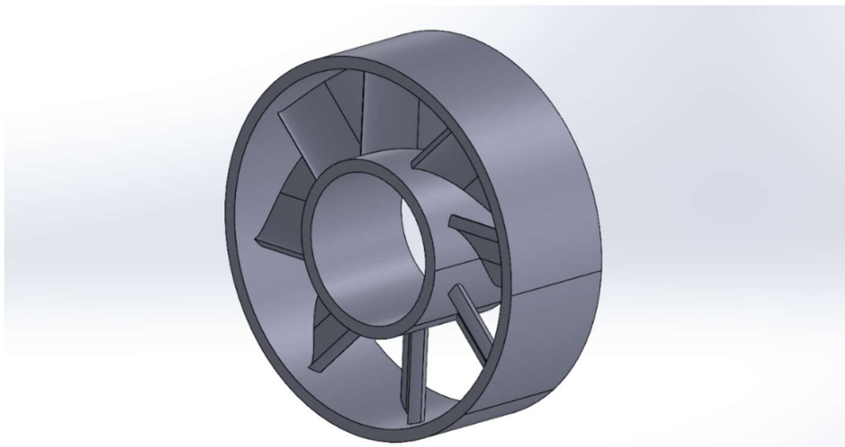

Fig 5.1 *Axial Curved Type Swirler.*

It features radial vanes connecting an inner hub and outer ring, creating a stable, uniform swirl pattern for efficient mixing and combustion.

5.2 AXIAL FLAT TYPE SWIRLER:

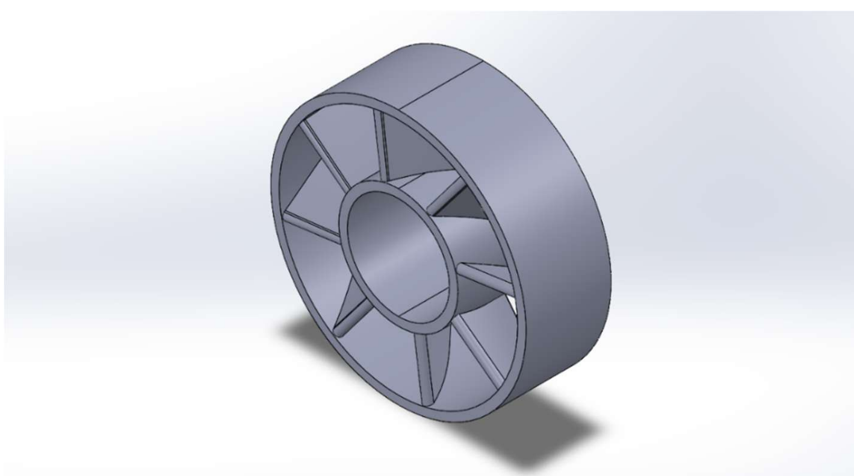

Fig 5.2 *Axial Flat Type Swirler.*

It consisting of an outer ring, inner hub, and evenly spaced vanes to guide airflow. The geometry is designed to generate controlled swirl for efficient air–fuel mixing in applications like gas turbine combustors.

5.3 RADIAL TYPE SWIRLER:

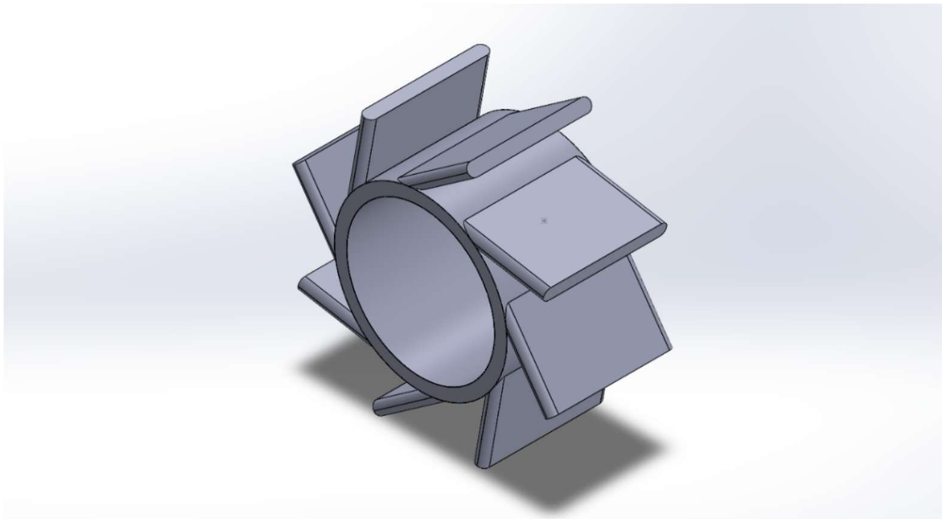

Fig 5.3 *Radial Type Swirler.*

This is a 3D CAD model of a radial swirler, featuring flat, outward-projecting vanes arranged around a central cylindrical hub.
The design redirects airflow outward at an angle to create strong radial turbulence and improved mixing in combustion or fluid-flow applications.

5.4 RADIAL TYPE SWIRLER WITH COMBUSTION CHAMBER

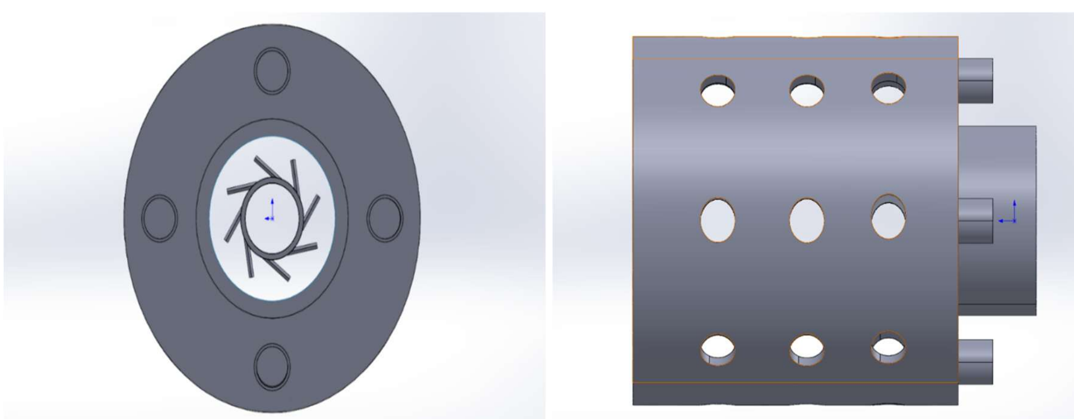

Fig 5.4(a) *Front and Side Radial Type Swirler with Combustion Chamber.*

This image shows a front view of a circular mounting plate with four equally spaced bolt holes and a central opening containing angled swirler vanes.The design indicates a swirler integrated into a flange assembly, typically used to control airflow entering a combustion or mixing system. side view of a perforated combustor liner, featuring multiple evenly spaced airflow holes arranged in rows.

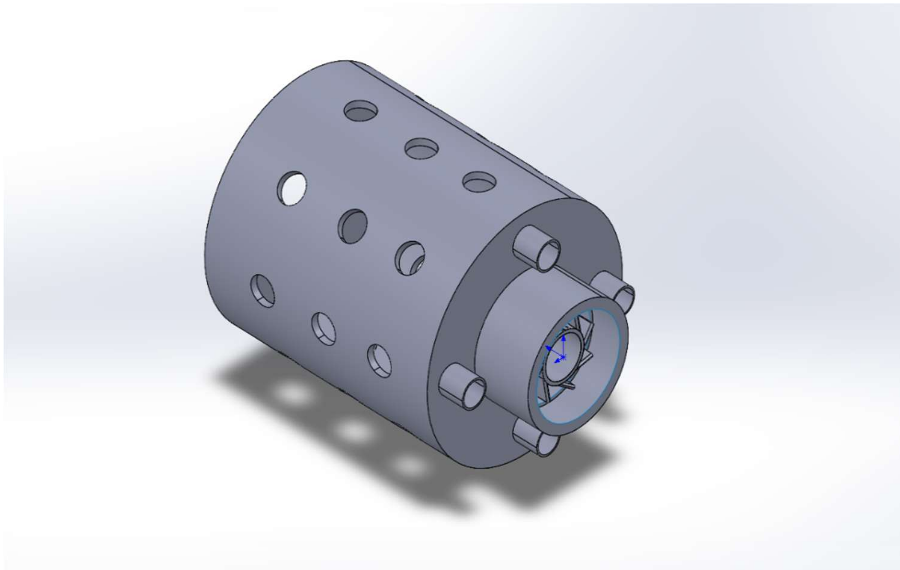

Fig 5.4(b) *Isometric View Radial Type Swirler with Combustion Chamber.*

This image shows a perforated cylindrical combustor liner with multiple airflow holes and a front-mounted swirler assembly secured by four bolts. The design allows primary air entry through the swirler and secondary air through the liner holes, ensuring effective mixing and stable combustion.

5.5 AXIAL CURVED TYPE WITH COMBUSTION CHAMBER

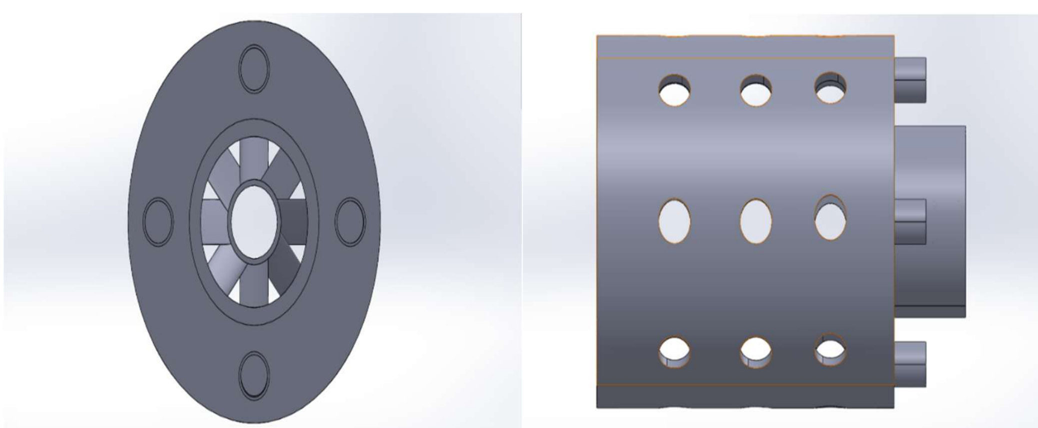

Fig 5.5(a) *Front and Side Axial Curved Type with Combustion Chamber.*

This image shows a front view of a flange-mounted axial swirler, with multiple curved vanes surrounding a central airflow passage. The four bolt holes around the perimeter indicate attachment to a combustor or duct, is a fuel injector.a side view of a perforated combustor liner, featuring multiple evenly spaced airflow holes arranged in rows. These holes allow secondary and dilution air to enter the combustion chamber, helping control temperature distribution and improve mixing efficiency.

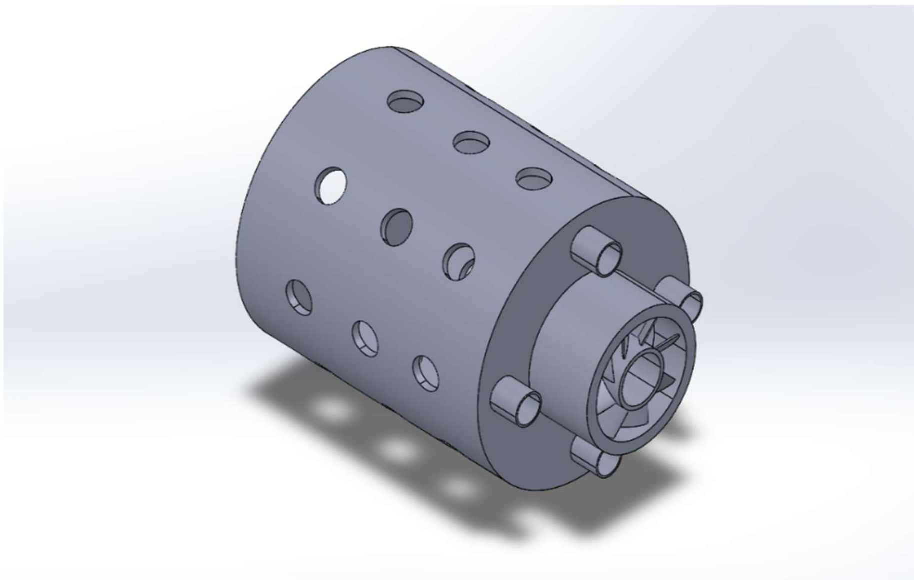

Fig 5.5(b) *Isometric view Axial Curved Type with Combustion Chamber.*

This image shows a 3D assembly of a combustor liner with multiple air-admission holes and a front-mounted axial swirler fixed using four bolts. The configuration enables primary swirling airflow through the swirler and secondary dilution airflow through the liner holes, supporting efficient and stable combustion.

5.6 AXIAL FLAT TYPE SWIRLER WITH COMBUSTION CHAMBER

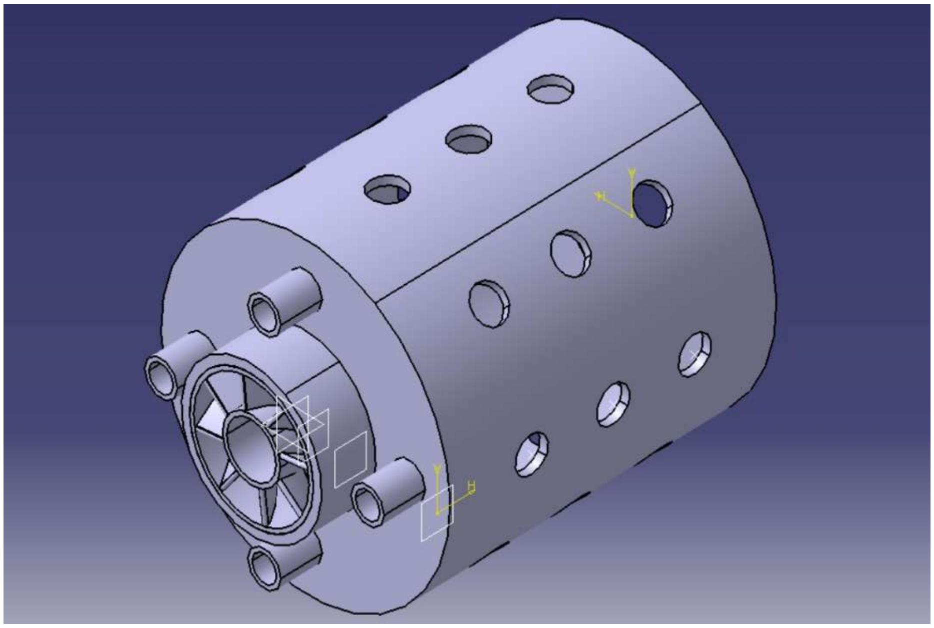

Fig 5.6(a) *Isometric view Axial Flat Type with Combustion Chamber.*

The presented model consists of an axial vane-type swirler integrated with a perforated cylindrical combustor liner. The swirler uses multiple fixed vanes arranged around a central hub to impart controlled rotational motion to the incoming primary air.

The surrounding cylindrical liner contains multiple rows of circular air-admission holes, which supply secondary and dilution air to control temperature distribution and improve combustion efficiency. In this configuration, the swirler handles the primary airflow responsible for initiating and anchoring the flame, while the liner holes manage secondary combustion and cooling. This design provides an optimal balance between swirl generation, pressure drop, and effective mixing, making it suitable for compact gas-turbine–type combustor applications.

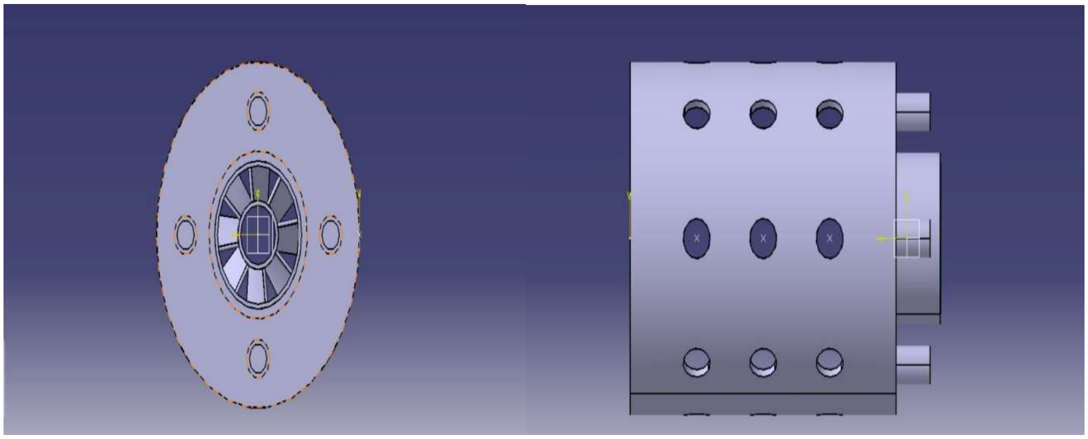

Fig 5.6(b) *Front and Side Axial Flat Type with Combustion Chamber.*

This model features an axial vane-type swirler mounted on a circular flange, with multiple straight vanes uniformly distributed around a central hub. The vane orientation forces the incoming air to rotate as it passes through the central passage, producing a controlled swirl required for initiating stable combustion. The flange includes four bolt holes, allowing the swirler to be rigidly attached to the combustor liner and ensuring precise alignment with the incoming airflow. side view of a perforated combustor liner, featuring multiple evenly spaced airflow holes arranged in rows.These holes allow secondary and dilution air to enter the combustion chamber, helping control temperature distribution and improve mixing efficiency.

CHAPTER 6

RESULTS AND OBSERVATIONS

6.1 CO_2 MASS FRACTION

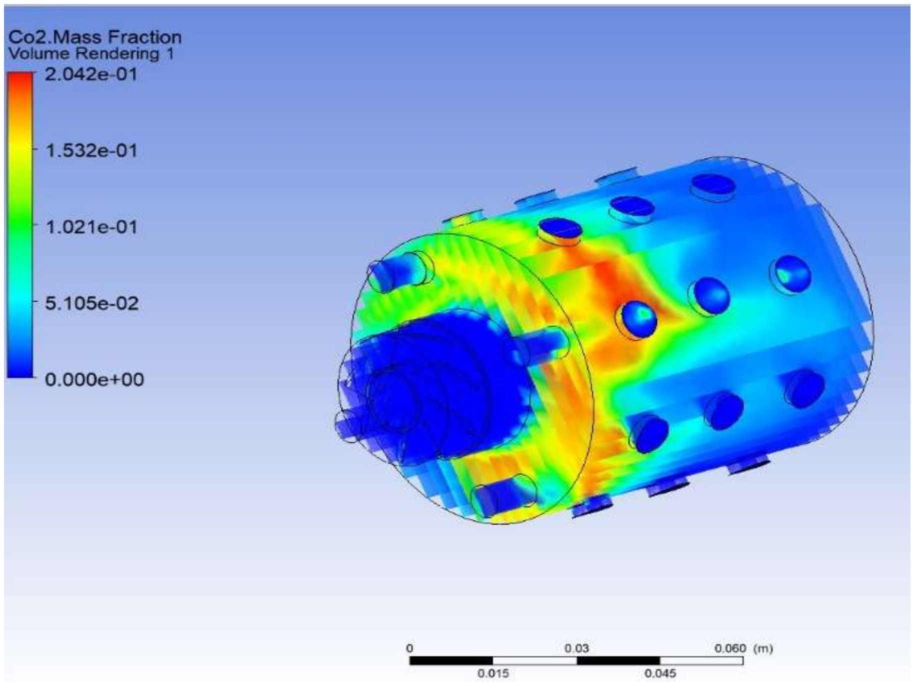

Fig 6.1 *Co_2 Mass Fraction Contour.*

The CFD result illustrates the CO_2 mass fraction distribution inside the swirler-integrated combustor liner. Higher CO_2 concentration regions (shown in yellow to red) indicate zones of more complete combustion, typically where fuel–air mixing is strongest. I Lower CO_2 regions (blue to green) on the sides and near the perforation holes represent zones dominated by secondary air penetration, which cools and dilutes the combustion products. The smooth transition of CO_2 concentration along the combustor length demonstrates effective secondary-air mixing through the liner holes. The overall distribution suggests that the swirler design promotes centralized combustion, uniform mixing, and efficient dilution, making the configuration suitable for compact gas-turbine combustor application.

6.2 O_2 MASS FRACTION

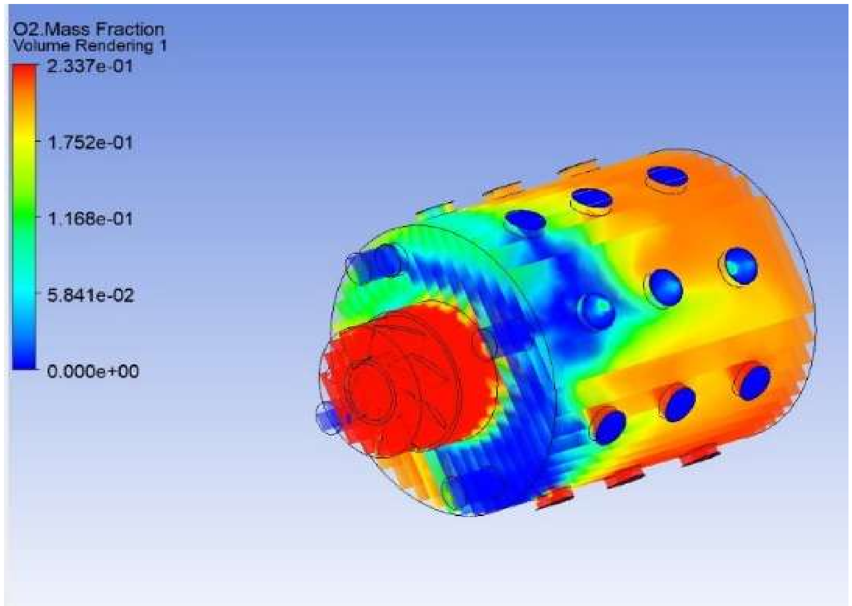

Fig 6.2 *O_2 Mass Fraction Contour.*

The O_2 mass fraction plot shows how oxygen is consumed and distributed throughout the combustor after passing through the swirler. The red region near the swirler inlet indicates that oxygen enters the chamber at a high concentration before mixing with fuel. As the flow moves downstream, oxygen levels drop sharply (seen as a shift from red/yellow to green/blue), confirming rapid combustion and effective fuel–air mixing immediately after the swirler.

Higher O_2 concentrations along the liner walls and near the dilution holes indicate fresh secondary air entering through the perforations, which contributes to cooling and completing the combustion process. The low-oxygen core region inside the combustor shows where the flame is stabilized and where most of the oxygen has been consumed. Overall, this distribution demonstrates that the swirler effectively drives strong central mixing, leading to efficient combustion while the liner holes supply controlled secondary air for temperature regulation.

6.3 PRESSURE CONTOUR

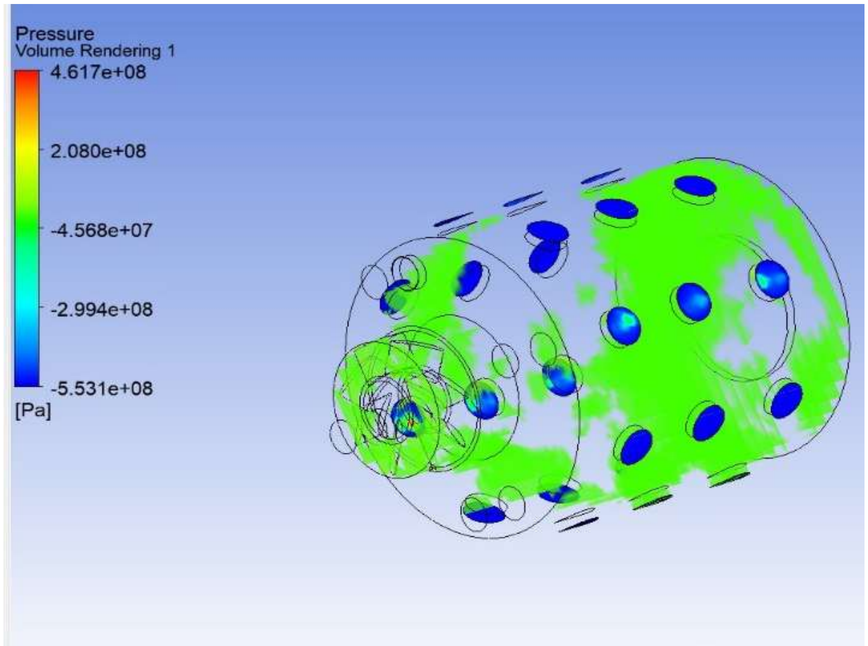

Fig 6.3 *Pressure Contour.*

The pressure contour illustrates how the airflow behaves inside the combustor after entering through the swirler. The red and yellow regions near the swirler face represent the highest-pressure zones, caused by flow stagnation and the resistance created by the swirler vanes. This confirms that the swirler generates a strong pressure gradient, which is essential for producing the necessary swirl intensity and driving the central recirculation zone.

As the flow moves downstream inside the combustor liner, the pressure gradually decreases (shown by green and blue shades). This reduction indicates accelerated flow, expansion, and mixing after passing through the restricted swirler opening. The moderate pressure variations around the liner perforation holes show where secondary air enters the chamber, creating localized disturbances that support additional mixing and cooling. Overall, the pressure distribution demonstrates that the swirler effectively induces controlled pressure drop and swirl, ensuring stable, well-mixed combustion throughout the liner.

6.4 N_2 MASS FRACTION

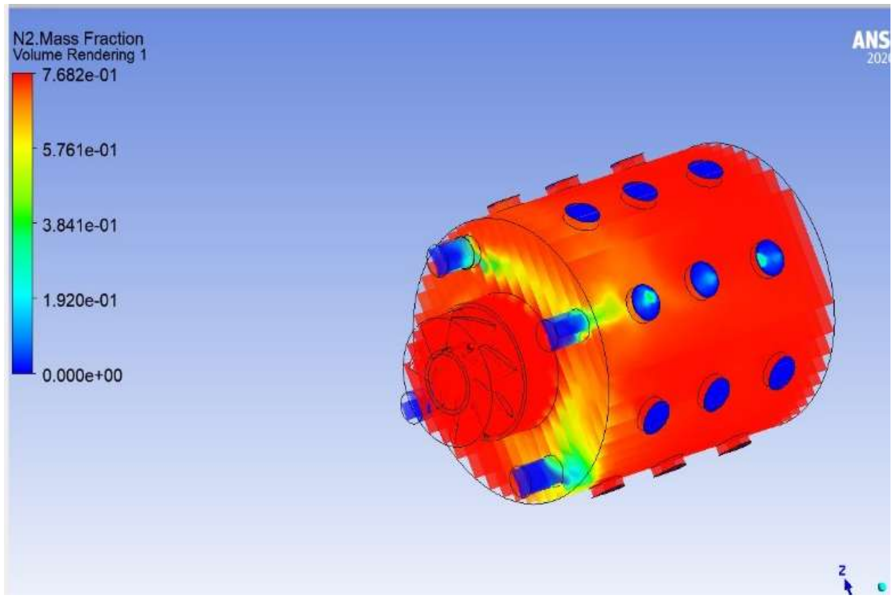

Fig 6.4 *N_2 Mass Fraction Contour.*

The N_2 mass fraction plot shows how nitrogen, which is inert in combustion, is distributed throughout the combustor during operation. The predominantly red coloration across the combustor liner indicates that nitrogen remains the major component of the mixture, as expected in air-based combustion systems. This high concentration confirms that nitrogen travels uniformly through the swirler and continues downstream with the combustion products.

Near the swirler region and around the liner holes, localized zones of reduced N_2 concentration (green to blue) appear where oxygen and fuel react to form combustion products such as CO_2 and H_2O. The secondary air entering through the perforated holes mixes with hot gases, restoring nitrogen concentration downstream. The overall distribution shows that the swirler effectively drives air into the combustor core, while the liner holes supply fresh air that blends smoothly with the combustion gases, maintaining consistent nitrogen levels throughout the chamber.

6.5 C12h23 MASS FRACTION

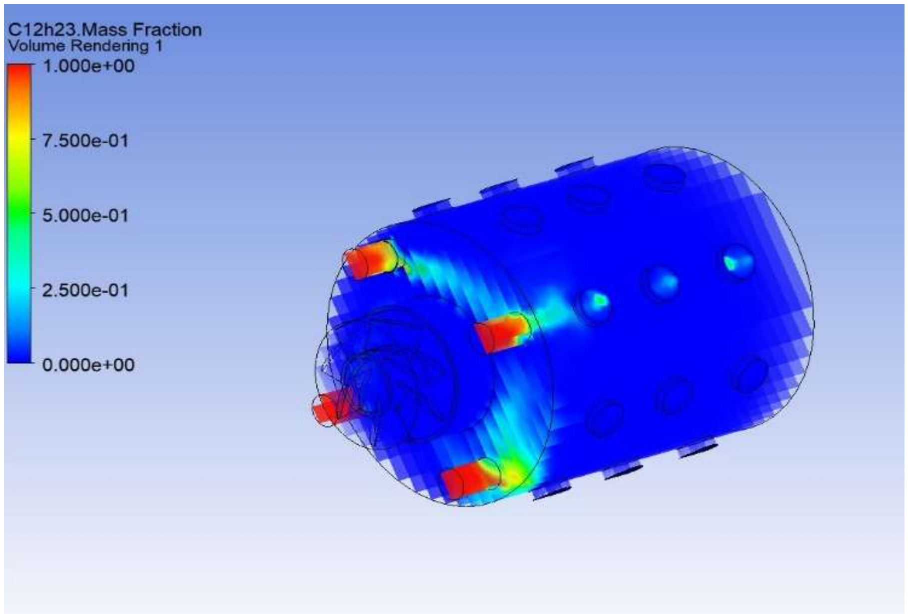

Fig 6.5 *$C_{12}h_{23}$ Mass Fraction Contour.*

C12h23 mass fraction contour represents the distribution of unburned fuel within the combustor. The highest fuel concentration (red zones) is located near the fuel injection region close to the swirler face, indicating where the liquid/air mixture first enters the chamber. As the flow moves downstream, the fuel mass fraction rapidly decreases (transitioning from yellow → green → blue), showing that the swirler induces strong mixing, causing the fuel to evaporate and react quickly with oxygen.

Near-zero fuel levels (dark blue) dominate most of the combustor volume, demonstrating efficient combustion with minimal unburned hydrocarbons. Slight localized fuel traces near the dilution holes suggest fresh air interaction causing minor disturbances in the mixing pattern, which is normal in perforated-liner designs. Overall, the distribution confirms that the axial swirler effectively disperses and mixes the fuel, ensuring rapid ignition and complete combustion inside the combustor liner

6.6 TEMPERATURE CONTOUR

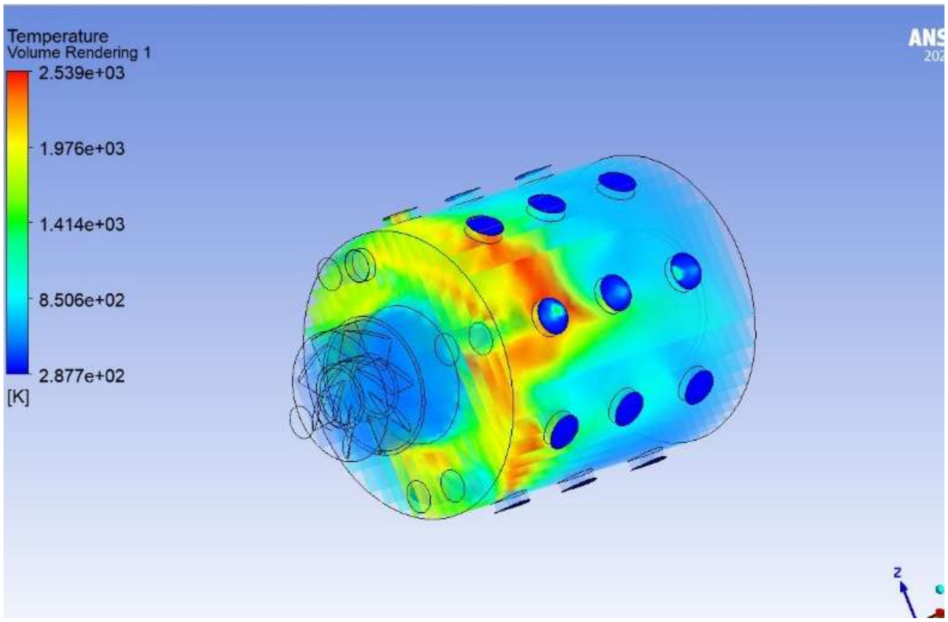

Fig 6.6 *Temperature Contour.*

The temperature contour shows how heat is generated and distributed inside the combustor after the airflow passes through the swirler. The highest-temperature region (yellow to red zone) appears just downstream of the swirler, indicating the primary combustion zone where the fuel–air mixture ignites and burns most intensely. This confirms that the swirler effectively creates a strong central recirculation zone (CRZ) that traps hot gases and stabilizes the flame.

As the flow moves further into the combustor liner, the temperature decreases smoothly (transitioning to green and blue). This reduction is caused by secondary air entering through the liner holes, which cools and dilutes the hot combustion gases. The temperature distribution across the liner shows no extreme hotspots along the wall, indicating efficient mixing and effective thermal management. Overall, the temperature field validates that the swirler–liner combination promotes stable combustion, uniform heat release, and good dilution characteristics inside the combustor.

6.7 VELOCITY CONTOUR

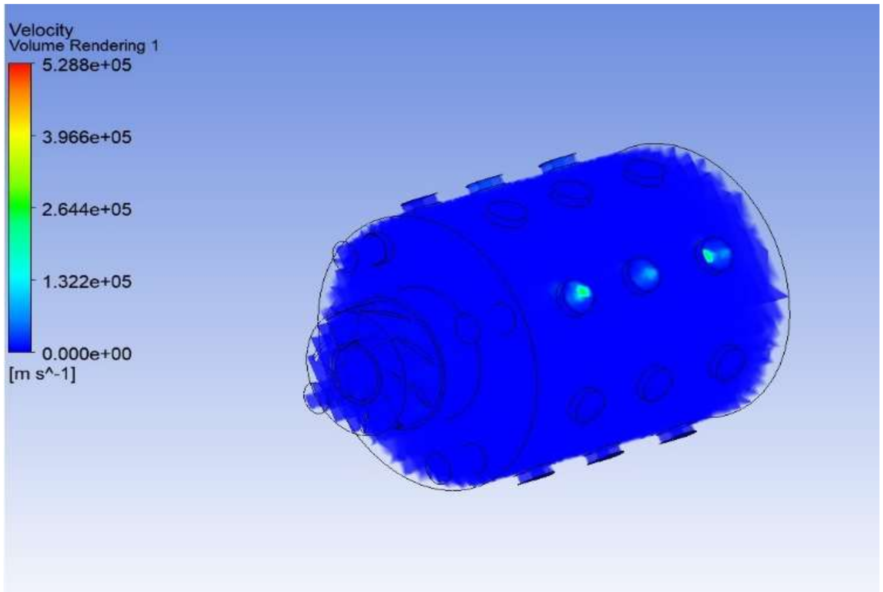

Fig 6.7 *Velocity Contour.*

The velocity contour shows how the flow accelerates and redistributes as it moves through the swirler and into the combustor liner. The highest velocity regions (yellow to green) appear near the dilution holes on the liner, indicating that secondary air jets enter the combustor at high speed. These jets enhance mixing and help cool the hot combustion gases downstream.

Near the swirler face, the flow velocity is relatively low (blue zones), showing the effect of flow deceleration and redirection as the air passes through the swirler vanes. This slowdown is essential for generating the central recirculation zone that stabilizes combustion. As the flow progresses, the velocity remains lower in the combustor core, where combustion occurs, and increases sharply again near the walls where secondary air injection takes place. Overall, the velocity field confirms that the swirler induces controlled flow deceleration and swirl, while the liner holes create high-velocity jets that improve mixing, flame stability, and temperature uniformity.

6.8 STATIC TEMPERTATURE GRAPH

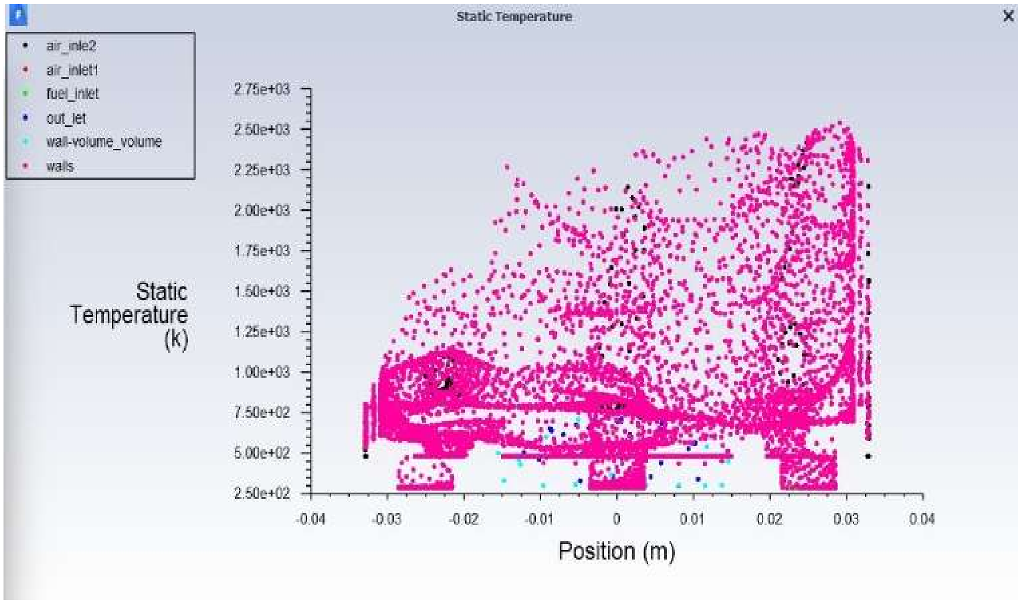

Fig 6.8 *Static Temperature Graph.*

The static temperature plot shows how temperature varies along the length of the combustor for different fluid zones. The cluster of high-temperature points (around 1500–2200 K) occurs immediately downstream of the swirler, indicating the primary combustion region where fuel and air rapidly mix and ignite. This confirms that the swirler effectively establishes a strong recirculation zone, sustaining stable and intense combustion close to the inlet.

As the position increases toward the outlet, the temperature gradually decreases due to secondary air entering through the liner holes, which cools and dilutes the hot gases. The lower-temperature scatter near the walls and outlet region shows effective heat distribution and cooling within the combustor. Overall, the plot demonstrates that the temperature peaks near the swirler—where combustion is strongest—and decreases progressively down the combustor, validating efficient mixing, thermal dilution, and stable combustion behaviour in this swirler–liner configuration.

6.9 STATIC PRESSURE GRAPH

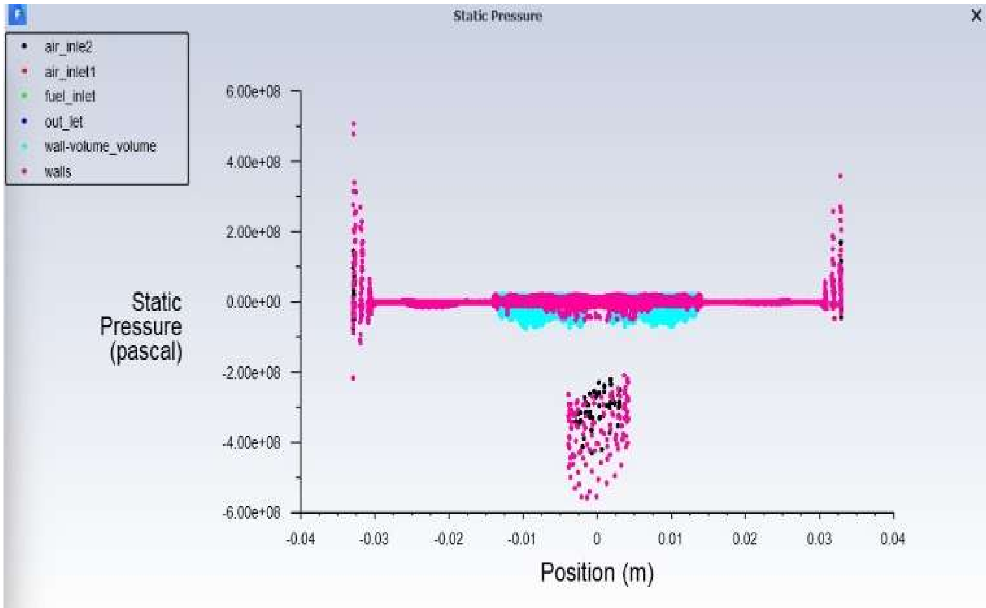

Fig 6.9 *Static Pressure Graph.*

The static pressure plot shows how pressure varies along the length of the combustor for different flow regions. The higher-pressure points near the inlet (around 3–6 × 10^8 Pa) represent the air and fuel entry sections, where the swirler creates a strong resistance to the incoming flow. This confirms that the swirler induces the expected pressure drop required to generate swirl and form the central recirculation zone.

As the position increases toward the combustor outlet, the pressure levels flatten and stabilize, indicating a gradual expansion and smooth flow progression inside the liner. The cluster of lower-pressure points around the central region corresponds to combustion-induced acceleration and secondary-air penetration through the liner holes. Overall, the pressure distribution shows an expected trend: high pressure before the swirler, a sharp drop across it, and a stable pressure field along the combustor, confirming that the swirler–liner configuration performs effectively in maintaining flow stability and driving efficient combustion.

6.10 Key Outputs

- Pressure and temperature contours (axial, tangential components).
- CO_2, 0_2, N_2, $C_{12}H_{23}$ Mass fractions over swirler.
- Recirculation zone visualization.

CHAPTER 7

CONCLUSION

The present works in this paper mainly focus on the procedure of designing axial swirler, and further investigation on flow characteristic of swirling flow generated by vane axial swirler by help of computational methods. Moreover, the effects of geometric parameters, namely vane angle and vane number and flow are also reported in the study.

Swirler design is carried out on the basis of semi-empirical equations. The design procedure using pressure drop method include three steps namely, mass flow calculation through air distribution, effective area calculation through pressure drops and sizing from the geometric relationship. The non-reacting swirling flow can be simulated very well through using k–ε realizable turbulence model provided in FLUENT that has been validated in this study.

Axial reverse flow velocity, turbulence along axial direction and pressure drop increases with the increase in angle of vanes. Namely the better fuel/air mixing and recirculation structure are obtained if the vane angle of swirler is larger but at the penalty of the pressure drop of combustors. Increasing the vane number increases the turbulence energy and decreases the axial flow velocity. For a given swirler, increasing mass flow at inlet decreases the pressure drop through the swirler and the pressure drop seems to keep constant if the mass flow is increased above a critical value. Thus, the mass flow through the swirler should be adjusted carefully when designing swirler. The results show that the axial velocity and radial velocity profiles have four different types of profiles.

CHAPTER 8

REFRENCES

[1] B. Khandelwal, D. Lili, and V. Sethi, “Design and study on performance of axial swirler for annular combustor by changing different design parameters,” Journal of the Energy Institute, vol. 87, no. 4, pp. 372–382, Nov. 2014, doi: 10.1016/j.joei.2013.07.006.

[2] N. A. Samiran, K. F. Abu Mansor, R. Abd Rashid, and M. S. S. Hamid, “CFD simulation analysis of non-premixed combustion using a novel axial–radial combined swirler for emission reduction enhancement,” CFD Letters, vol. 17, no. 7, pp. 18–31, 2023, doi: 10.37934/cfdl.15.6.111.

[3] D. A. Zavaleta-Luna, M. O. Vigueras-Zúñiga, A. L. Herrera-May, S. A. Zamora-Castro, and M. E. Tejeda-del-Cueto, “Optimized design of a swirler for a combustion chamber of non-premixed flame using genetic algorithms,” Energies, vol. 13, no. 9, Art. no. 2240, 2020, doi: 10.3390/en13092240.

[4] J. Lin, M. Bao, F. Zhang, Y. Zhang, and J. Yang, “Numerical and experimental investigation of a non-premixed double swirl combustor,” Energies, vol. 15, no. 2, Art. no. 458, 2022, doi: 10.3390/en15020458.

[5] H.-J. Ju, J. H. Cho, and J. Hwang, “Numerical analysis of flow characteristics of a swirl-stabilized premixed burner with different swirl vane configurations,” Journal of Mechanical Science and Technology, vol. 36, pp. 5793–5800, Nov. 2022, doi: 10.1007/s12206-022-1040-7.

[6] M. Kumari, “Experimental analysis of flow through rotating combustion swirler with zero-degree inlet and outlet angle of guide vane,” International Journal of Computer Science & Engineering Technology (IJCSET), vol. 5, no. 8, pp. 820–824, Aug. 2014.

[7] K. Gupta, D. G. Lilley, and N. Syred, Swirl Flows. New York, NY, USA: Springer, 1984.

[8] H. Lefebvre and D. R. Ballal, Gas Turbine Combustion: Alternative Fuels and Emissions, 3rd ed. Boca Raton, FL, USA: CRC Press, 2010.

[9] T. Poinsot and D. Veynante, Theoretical and Numerical Combustion, 3rd ed. Philadelphia, PA, USA: Edwards Publishing, 2011.

[10] V. N. Reddy, A. K. Agarwal, and S. K. Aggarwal, “Effect of swirler design on flame stabilization and emissions in gas turbine combustors,” Fuel, vol. 238, pp. 312–323, Jan. 2019, doi: 10.1016/j.fuel.2018.10.111.

[11] M. Streibel and K. H. Becker, “Impact of swirl intensity on flame stability and combustion efficiency for non-premixed flames,” Combustion and Flame, vol. 162, no. 10, pp. 3728–3739, Oct. 2015, doi: 10.1016/j.combustflame.2015.06.018.

[12] S. G. Tuttle, R. Seitzman, and B. T. Zinn, “Combustion dynamics of swirl-stabilized flames,” Journal of Engineering for Gas Turbines and Power, vol. 132, no. 11, Art. no. 111502, Nov. 2010, doi: 10.1115/1.4000575.

[13] H. Sholeikhah, M. Saidi, and M. R. Malayer, “Investigation of vane angle and swirl number effects on gas-turbine combustor aerodynamic performance,” Aerospace Science and Technology, vol. 72, pp. 329–338, Oct. 2017, doi: 10.1016/j.ast.2017.01.006.

[14] R. K. Singh and S. Kumar, “CFD analysis of combustor swirlers with different vane geometries for improved mixing and low emissions,” International Journal of Turbo & Jet Engines, vol. 39, no. 4, pp. 423–437, 2022, doi: 10.1515/tjj-2020-0094.

[15] C. Fernandez, J. D. Kittelson, and D. L. Reuss, “Effect of swirl ratio and injection timing on flame structure in diesel combustion,” SAE Technical Paper 2002-01-1106, 2002, doi: 10.4271/2002-01-1106.

[16] X. Li, K. Yu, and Y. Xu, “Study of non-premixed combustion in a swirler-stabilized burner under varying equivalence ratios using ANSYS Fluent,” Applied Thermal Engineering, vol. 195, Art. no. 117201, Apr. 2021, doi: 10.1016/j.applthermaleng.2021.117201.

[17] M. A. Habib, M. S. Sayeed, and M. A. Badruddin, “Effect of swirl intensity on pollutant formation and flame temperature in gas-turbine combustors,” Energy Conversion and Management, vol. 49, no. 12, pp. 3379–3388, Dec. 2008, doi: 10.1016/j.enconman.2008.05.029.

[18] R. Shadle, D. V. Dupont, and J. R. Wilson, “Swirling flow characteristics of gas turbine combustor air passages,” Journal of Propulsion and Power, vol. 22, no. 3, pp. 670–678, May 2006, doi: 10.2514/1.16408.

[19] Sadiki, R. Janicka, and A. Kempf, “Large eddy simulation of turbulent swirling non-premixed combustion,” Flow, Turbulence and Combustion, vol. 79, pp. 1–28, 2007, doi: 10.1007/s10494-007-9094-3.

[20] J. F. Driscoll, “Turbulent premixed combustion: Flamelet structure and flame stability,” Progress in Energy and Combustion Science, vol. 34, no. 1, pp. 91–134, Feb. 2008, doi: 10.1016/j.pecs.2007.05.002.

[21] L. Burrus and M. Matalanis, “Fuel–air mixing enhancement using counter-rotating swirlers,” AIAA Journal, vol. 46, no. 6, pp. 1475–1484, Jun. 2008, doi: 10.2514/1.32341.

[22] S. Kashir and S. Hossain, “CFD simulation of axial swirl injectors for low-emission combustors using k–ω turbulence model,” Aerospace Science and Technology, vol. 85, pp. 172–183, Sep. 2019, doi: 10.1016/j.ast.2018.11.033.

[23] G. M. Razdan and R. S. Pannu, “Experimental investigation of combustion stability in a gas turbine combustor with varying swirler vane angles,” ASME Journal of Engineering for Gas Turbines and Power, vol. 141, no. 9, Art. no. 091020, Sep. 2019, doi: 10.1115/1.4044023.

[24] S. Menon and R. G. Sankaran, “LES-based modelling of turbulent combustion with recirculation zones and swirl effects,” Combustion Science and Technology, vol. 175, no. 8, pp. 1379–1398, Aug. 2003, doi: 10.1080/00102200302394.

[25] T. Claridge, P. Stief, and A. Dreizler, “Laser diagnostics investigation of flame stabilization in a swirl-stabilized non-premixed combustor,” Combustion and Flame, vol. 161, no. 12, pp. 3100–3113, Dec. 2014, doi: 10.1016/j.combustflame.2014.06.020.